

Biology

Sem 1

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===== SOURCE_FILE: nexus/Biology/Notes/Coming Soon.url =====

CATEGORY: Notes

STATUS: url_reference_only

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===== SOURCE_FILE: nexus/Biology/PYQs/PYQ Dec 2024.pdf =====

CATEGORY: PYQs

STATUS: ok

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Reg. No.

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2024

First Semester

21BTB103T-BIOLOGY

(For the candidates admitted from the academic year 2024 - 2025)

Note:

(i) Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall

invigilator at the end of 40th minute.

(ii) Part - B & Part - C should be answered in answer booklet.

Time: 3 hours Max. Marks: 75

PART - A (20 X 1 = 20 Marks)

Answer ALL Questions

Marks BL CO PO

1. DNA in prokaryotes are present in

A) Mitochondria B) Ribosomes

C) Plasma membrane D) Nucleoid region

1 2 1 1

2. How many chromosomes are present in human cell?

A) 46 pairs B) 46

C) 23 D) 12 pairs

1 2 1 1

3. Calcium ion regulation is done by which part of the cell

A) plasma membrane B) cytoplasm

C) golgi complex D) endoplasmic reticulum

1 2 1 I

4. Covalent bond that is present in protein that joins aminoacids is

- A) peptide bond B) disulphide bond
- C) hydrogen bond D) glucosidic bond

1 2 2 1

5. Sugar in DNA molecule is

- A) Ribose B) Deoxyribulose
- C) Deoxyribose D) Glucose

1 2 2 1

6. The bond that connects the sugar molecules in starch is

- A) peptide bond B) glycosidic bond
- C) phospho diester bond D) hydrogen bond

1 2 2 1

7. Fatty acid is stored as-----for energy requirement

- A) fats B) phospho lipid
- C) wax D) glycerol

1 2 .2 1

8. Genetic information is carried from nucleus to cytoplasm through

- A) DNA B) RNA
- C) protein D) fibers

1 2 2 2

9. Penicilin kills bacteria by

- A) inhibiting cell wall synthesis B) by stoping replication
- C) by inhibiting its motility D) by inhibiting translation

1 2 3 1

10. Tetracyclin resistance to bacteria is by

- A) enzyme modification B) altered targets
- C) efflux pumps D) decreased uptake

1 2 3 1

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11. Tetanus Toxoid vaccine is a ——

- A) protein
- C) DNA of Closteridium tetani < .
- B) killed bacteria
- D) killed Virus

1 2 3 1

12. Antibiotics are used to treat

- A) Parasite infection
- C) Bacterial Infection
- B) Viral Infection
- D) Life style disease

1 2 3]

13. Enzyme used in glucose monitor is j_____

- A) glucose oxidase
- C) peroxidase
- B) sucrase
- D) catalase

1 2 4 1

14. Energy requirement for protein based machines is met by

- A) ATP B) proton motive force
- C) ADP D) NAD

1 2 4 1

15. Kinesin family of protein moves along

- A) Myosin
- C) Dynein
- B) Actin
- D) Microtubules

1 2 4 1

16. Stents used for angioplasty are made of

- A) Titanium
- C) Composite
- B) Aluminium
- D) Ceramic

1 2 5 1

17. Most commonly used biomaterial in dental application is

- A) Ceramic B) Metal
- C) Polymer D) Alloy

1 2 5 1

18. Which of the following can mimic the teeth

- A) Hydroxyapatite
- C) Magnesium
- B) Calcium
- D) Iron

1 2 5 1

19. Which of the following natural polymer is used as biomaterial

- A) Starch B) Glycogen
- C) Chitosan D) Potato

1 2 5 1

20. Which of the property is essential for biomaterial if it is used in bone replacement

- A) load bearing B) thermal conductivity
- C) elasticity D) opaque nature

1 2 5 1

PART - B (5 X 8 = 40 Marks) Marks BL CO PO

Answer ALL Questions

21 a. Explain the function of endoplasmic reticulum and its infraction with golgi complex in
' j

transport of macromolecules

(OR)

b. Discuss the current and potential future applications of stem cells in regenerative » 3 i
i
medicine
22 a. Discuss various classes of lipids and their functions in biological system 8221
(OR)
b. How do plants produce glucose by photosynthesis? Explain with diagrams . « 221
23 a. Differentiate attenuated vaccines from subunit vaccines on their mechanism of action
■ ® 331
(OR)
b. Discuss the mechanism of bacterial resistance against antibiotics. 8331
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24 a. Discuss the various types of biosensors with its mechanism 8 3 4 2
(OR)
b. Explain how ATP is generated by ATP synthase with diagrams 342
25 a. How cargo is transported in the cells? Explain with diagrams 342
(OR)
b. Discuss the properties of biomaterials that are used in medical field. 8 3 5 2
PART - C (1 X 15 = 15 Marks) Marks BL co PO
Answer ANY ONE Question
26. Explain the use of stem cells in medical application and elaborate the ethical issue 412
associated with it
27. Bacteria attain resistance due to overuse of antibiotics. What are all the risk
associated 4 3 2
with this? How this can be addressed?

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